

# Duong H. Le

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## Education

### Vietnam National University - Ho Chi Minh city, University of Technology

Ho Chi Minh City, Vietnam

B.ENG. IN COMPUTER SCIENCE (HONORS)

Aug. 2016 - Nov. 2020

- Cumulative GPA: 8.49/10 (Ranked 9/230)
- Thesis: Toward real-world High-Performance Computing Cluster Job Scheduling with Reinforcement Learning (Advisor: Prof. Nam Thoai)

## Research Interests

My research interests focus on learning with limited supervision in the computer vision domain.

## Experience

### Allen Institute for AI

PREDOCTORAL RESEARCHER

Seattle, WA

Sep. 2022 - PRESENT

Supervisor: Dr. Jiasen Lu

- Conduct research on text to 3D models with the large-scale paired dataset of shapes and language.

### Stanford University

Stanford, CA (remote)

RESEARCH INTERN

Jun 2022 - Nov 2022

Supervisor: Prof. Leonidas Guibas

- Work closely with Mikaela Uy on manipulating and editing neural fields.
- Propose an algorithm for shape deformation on the implicit fields via a flow-based network. The introduced approach faithfully preserves geometric and appearance details and can be used to edit radiance fields trained on 3D assets from Sketchfab, Blendswap.

### VinAI Research

Ho Chi Minh City, Vietnam

AI RESIDENT

Jul. 2020 - Sep. 2022

Supervisor: Dr. Binh-Son Hua

- Conducting research on Machine Learning, Computer Vision and publishing paper in top-tier conferences.
- *Network Pruning that Matters*: investigate the importance of detail that is often unstated (i.e., learning rate schedule) to demonstrate the difficulty of attributing performance gain in network pruning. Recommend a better practice for comparing pruning algorithms, and introduce an effective fine-tuning scheme as a by-product.
- *Improving Few-shot Learning*: propose, conduct experiments and write the article for the idea of using distractor samples to bolster few-shot learning models that is effective and efficient in several evaluation protocols and standard datasets.
- *Set discrimination for Self-supervised Learning*: propose the idea of leveraging set-structured learning for self-supervised learning, implement, conduct experiments, and write the article.

### AdaComp Lab, National University of Singapore

Singapore, Singapore

RESEARCH INTERN

Jun. 2019 - Sep. 2019

Supervisor: Prof. David Hsu

- *GoAnywhere@NUS*: Learn about Imitation Learning, Path Planning and Robotic Operation System (ROS). Set up experiments with Pioneer 3-DX robot, conduct studies with variants of Intention-Net to leverage new sensors for automatic robot navigation, collect data for imitation learning.

### Nakagawa Lab, Tokyo University of Agriculture and Technology

Tokyo, Japan

RESEARCH INTERN

Jun. 2018 - Aug. 2018

Supervisor: Prof. Masaki Nakagawa

- *Apply GAN to improve old Japanese characters recognition*: Implement generative networks (GAN, VAE) to augment the dataset of old Japanese characters' images and train a classifier with augmented data with Tensorflow/Keras.

### HPC Lab, Ho Chi Minh City University of Technology

Ho Chi Minh City, Vietnam

RESEARCH ASSISTANT

Mar. 2018 - Aug. 2020

Supervisor: Prof. Nam Thoai

- Learn about Parallel Computing, scheduling systems
- Propose, implement the idea of utilizing knowledge distillation and ensemble to improve the efficacy of network pruning (BMVC'20).
- Discuss, clarify the idea of using Transformer and leverage the self-attention to predict popularity of online contents for caching and write article (HPCC'19).

## Publications

\* indicates equal contribution.

## Neural Implicit Deformer: Shape-to-Shape Deformation via Neural Flows

*under review*

DUONG H. LE, MIKAELA UY, IAN HUANG, DESPOINA PASCHALIDOU, ANIRUDDHA KEMHAVI, BINH-SON HUA, LEONIDAS GUIBAS

## MIRACLE: Multiple-Instance Random Sets for Contrastive Learning

*under review*

DUONG H. LE, BINH-SON HUA

## POODLE: Improving Few-shot Learning via Penalizing Out-of-Distribution Samples

*NeurIPS'21*

DUONG H. LE\*, KHOI D. NGUYEN\*, KHOI NGUYEN, QUOC-HUY TRAN, RANG NGUYEN, BINH-SON HUA

## Network Pruning that Matters: A Case Study on Retraining Variants

*ICLR'21*

DUONG H. LE, BINH-SON HUA

## Paying more Attention to Snapshots of Iterative Pruning: Improving Model Compression via Ensemble Distillation

*BMVC'20*

DUONG H. LE, TRUNG-NHAN VO, NAM THOAI

## Attention-based Neural Network: A Novel Approach for Predicting the Popularity of Online Content

*HPCC'19*

MINH-TRI NGUYEN, DUONG H. LE, TAKUMA NAKAJIMA, MASATO YOSHIMI, NAM THOAI

## Technical Skills

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**Language** Python, JavaScript  
**Framework** Pytorch, Jax, Numpy, Pandas, Matplotlib, Seaborn  
**Developer Tool** Git, Docker, LaTeX

## Honors & Awards

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2019 **2nd**, Digital Race 2019 University Round - Self-driving contest for Undergraduate Students *Ho Chi Minh City*  
2018 **Awardee**, JASSO Scholarship for Exchange Students *Tokyo*  
2016-2019 **Awardee**, HCMUT Scholarship for Outstanding (Top 3%) Students of Class *Ho Chi Minh City*

## Community Service

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2022 **Reviewer**, AAAI 2023, CVPR 2023

## Certifications

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Nov. 2021 **TOEFL ibt**, R:27, L:23, W:23, S: 21 *94/120*

## Reference

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### Dr. Jiasen Lu

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### Dr. Binh-Son Hua

RESEARCH SCIENTIST, VINAI RESEARCH

*v.sonhb@vinai.io*

### Dr. Quoc-Huy Tran

CTO, RETROCAUSAL AI

*huy@retrocausal.ai*